

Cymatic K-Space Mechanics

Complete Mathematical Framework

Registry: [\[CKS-MATH-0-2026\]](#)

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Parent Framework: [\[CKS-0-2026\]](#)

Logical Next Step: [\[CKS-MATH-1-2026\]](#) The Mechanical Necessity of Integer Quantization in Physical Systems

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Domain: Foundational Mathematics / Discrete Geometry

Status: CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [\[CKS-NEXT-1-2026\]](#)

Old Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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Axiomatic Foundation

Axiom 1 (Substrate Topology)

Graph: $G = (V, E)$

- 3-regular planar graph, Euler characteristic $\chi = 2$
- Nodes: $|V| = N = 3M^2$, $M \in \mathbb{N}$
- Edges: $|E| = 9M^2/2$
- Faces: $|F| = 3M^2/2 + 2$
- Coordination: $z = 3$ (every node has exactly 3 neighbors)

Construction (Three-Sector Rhombic Manifold):

- Take three $M \times M$ rhombic arrays from hexagonal lattice Λ
- Rotate by $2\pi s/3$ for $s \in \{0,1,2\}$
- Identify radial edges pairwise
- Result: Closed, boundary-free discrete 2-sphere
- Symmetry: Cyclic group C_3

Axiom 2 (Phase Dynamics)

State space: $\theta = (\theta_1, \dots, \theta_N) \in \mathbb{T}^N$

Evolution: Kuramoto coupling on graph G

$$d\theta_k/dt = \omega_k + \sum_{j \in N(k)} \beta_{\{kj\}} \sin(\theta_j - \theta_k)$$

Where:

- $\omega_k \in \mathbb{R}$ (natural frequency)
- $\beta_{\{kj\}} = \beta_{\{jk\}} > 0$ (symmetric coupling)
- $N(k) = \{3 \text{ nearest neighbors in graph } G\}$

Core Theorems

Theorem 1 (Topological Closure)

$$V - E + F = 3M^2 - 9M^2/2 + (3M^2/2 + 2) = 2 = \chi$$

Graph is homeomorphic to discrete 2-sphere.

No boundary. All nodes interior with $z = 3$.

Theorem 2 (Measure Preservation)

Flow is divergence-free: $\nabla \cdot (d\theta/dt) = 0$

Proof: For symmetric $\beta_{\{kj\}}$, each edge contributes:

$$\partial F_k / \partial \theta_k + \partial F_j / \partial \theta_j = -\beta \cos(\theta_j - \theta_k) + \beta \cos(\theta_j - \theta_k) = 0$$

$\therefore$ Uniform measure $d\mu = d\theta_1 \wedge \dots \wedge d\theta_N$ invariant (Liouville).

Theorem 3 (Gradient Flow Structure)

For uniform $\omega_k = \omega$:

$$\text{Potential: } V(\theta) = -\sum_{\{k,j\}} \beta_{\{kj\}} \cos(\theta_j - \theta_k)$$

$$\text{Evolution: } d\theta_k/dt = \omega - \partial V / \partial \theta_k$$

$$\text{Dissipation: } dV/dt = -\sum_k (d\theta_k/dt - \omega)^2 \leq 0$$

System minimizes frustration energy.

Theorem 4 (Synchronization Stability)

Synchronized state: $\theta_k = \omega t + \theta_0$ ($\forall k$)

Linear stability: Perturbation $\delta\theta$ evolves as

$$d(\delta\theta)/dt = L(\delta\theta)$$

where L = graph Laplacian with spectrum:

$$0 = \lambda_0 > \lambda_1 \geq \dots \geq \lambda_{N-1}$$

All non-zero modes decay: $e^{\lambda_i t} \rightarrow 0$

$\therefore$ Synchronized state asymptotically stable for all $\beta > 0$.

Theorem 5 (Spectral Gap)

For hexagonal lattice: $\lambda_1 \sim 1/M^2$

Mixing time: $\tau_{\text{mix}} \sim M^2$

Coherence time: $\tau_{\text{coh}} \sim M^2$

Synchronization rate: $\gamma = -\lambda_1 \sim 1/M^2$

Theorem 6 (Geometric Frustration)

Elementary triangle $\{k, j, m\}$:

Cannot simultaneously satisfy:

$$\theta_j - \theta_k = \alpha$$

$$\theta_m - \theta_j = \alpha$$

$$\theta_k - \theta_m = \alpha$$

Because: $\Sigma(\text{phase differences}) = 3\alpha \neq 0 \pmod{2\pi}$

$\therefore$ No global energy minimum exists.

$\therefore$ Rich phase structure beyond simple synchronization.

Coherence Scaling

Definition (Global Coherence)

$$C(M) = 1 - 1/(2M\sqrt{3}) = 1 - \sqrt{3}/(6M)$$

Properties:

- $C(1) = 1 - \sqrt{3}/6 \approx 0.711$
- $C(M) \rightarrow 1$ as $M \rightarrow \infty$

- $dC/dM = \sqrt{3}/(6M^2) > 0$ (monotonic)
- $C \sim 1 - O(M^{-1})$ for large M

Definition (Order Parameter)

Kuramoto order parameter:

$$Z(t) = (1/N) \sum_k e^{i\theta_k(t)} = r(t) e^{i\psi(t)}$$

where:

- $r \in [0,1]$: coherence magnitude
- $\psi \in [0, 2\pi]$: mean phase

Bounds:

- $r = 0 \Leftrightarrow$ uniform distribution
 - $r = 1 \Leftrightarrow$ perfect synchronization
-

Discrete Scale Invariance

Theorem 7 (Hierarchical Scaling)

$$N(kM) = 3(kM)^2 = k^2 N(M)$$

$$\text{Doubling: } N(2M) = 4N(M)$$

$$\text{Tripling: } N(3M) = 9N(M)$$

Enables block-spin renormalization:

- Coarse-grain $k \times k$ blocks
 - Effective coupling: $\beta_{\text{eff}} \sim k^2\beta$
 - Hierarchy preserved
-

Special Solutions

S1: Uniform State

$$\theta_k = \omega t + \theta_0 \text{ (all phases equal)}$$

$$\text{Condition: } \omega_k = \omega \forall k$$

Stability: Asymptotically stable $\forall \beta > 0$

Basin: Almost all initial conditions (generic)

S2: Three-Sector State

Sector 0: $\theta_k = \omega t$

Sector 1: $\theta_k = \omega t + 2\pi/3$

Sector 2: $\theta_k = \omega t + 4\pi/3$

Respects C_3 symmetry.

Equilibrium for balanced neighbor counts.

Stability: Saddle point (unstable manifold exists).

S3: Spiral Waves

$\theta_k(t) = \omega t + \Phi(r_k, \varphi_k)$

where $\Phi(r, \varphi) = q\varphi + f(r)$

q = topological charge (winding number)

Existence: Numerical evidence for $\beta \in [\beta_{\min}, \beta_{\max}]$

Stability: Stable for intermediate coupling

Analytical proof: Open problem

Critical K-Space Constraint

! DO NOT FOURIER TRANSFORM TO REAL SPACE

Five Fundamental Traps:

1. Non-Local Paradox

Coupling $\beta_{\{kj\}}$ defined on abstract graph adjacency.

NOT derived from x-space potential $U(r)$.

Consequence: No local kernel exists.

In x-space: Interactions appear non-local.

Stay in k-space to preserve locality.

2. Nyquist-Shannon Violation

$N = 3M^2$ is topological constraint on spectrum itself.

θ_k are PRIMARY degrees of freedom.

NOT Fourier coefficients of x-field.

Trap: Interpolating with sinc/Lanczos kernels.

Result: Breaks node count, destroys C_3 symmetry.

3. Origin Singularity

$k = 0$ is conical pole where three sectors meet.

NOT a trivial DC offset.

Standard FFT assumes:

- Rectangular grid
- Periodic BC (toroidal, $\chi = 0$)

Reality:

- Spherical topology ($\chi = 2$)
- Radial edge identifications

Applying FFT: Hallucinates ghost reflections.

4. Phase vs. Group Velocity

$d\theta_k/dt$ = internal phase rotation (S^1 dynamics)

NOT spatial velocity dx/dt

Misinterpretation: θ as translational speed

Consequence: Apparent FTL propagation

Reality: No Lorentz invariance asserted

5. Sphere vs. Torus

This framework: $\chi = 2$ (discrete 2-sphere)

Standard FFT: $\chi = 0$ (periodic torus)

Toroidal BC would require:

$V - E + F = 0$ (contradicts Theorem 1)

Forcing torus breaks:

- $N = 3M^2$ closure
- $C(M)$ coherence formula
- Spectral gap scaling

Prescription for Practitioners:

- ✓ Treat G as abstract graph
- ✓ Distance = graph geodesic (edge count)
- ✓ Compute entirely in k -space
- ✓ Visualize final state only
- × Do NOT map to $\mathbb{R}^2$ grid
- × Do NOT use standard FFT

- × Do NOT impose periodic BC
 - × Do NOT interpolate between k-nodes
-

Numerical Implementation

Stability Criterion

Euler method: $dt < 2/(3\beta)$

RK4 method: $dt < 4/(3\beta)$ (empirical)

Complexity: $O(M^2 T/dt)$

Parallelization: Embarrassingly parallel (GPU-friendly)

The 10 Inviolable Operational Rules in CKS

$K \rightarrow X$ and beyond

1. $K \rightarrow X$ Rule

Never inverse-Fourier; only **interference summation** $\psi(x) = \sum_k \varphi_k e^{ik \cdot x}$ is permitted.

2. Adjacency Rule

Distance $\equiv$ graph-hop count; no $\mathbb{R}^2$ metric is ever introduced.

3. Coordination Rule

Every node keeps $z=3$; no boundary, no dangling edges.

4. Parameter Rule

Only two inputs: integer M and real $\beta > 0$; no tunable constants.

5. Closure Rule

Node count must satisfy $N=3M^2$ exactly; any other N breaks $\chi=2$.

6. Symmetry Rule

The graph carries an exact C_3 rotation; all modes fall into χ_0, χ_1, χ_2 irreps.

7. Gradient Rule

For uniform ω , the flow is ∇V with $dV/dt \leq 0$; chaos is impossible.

8. Coherence Rule

Global coherence $C(M) = 1 - 1/(2M\sqrt{3})$ is monotonic and parameter-free.

9. Frustration Rule

Elementary triangles forbid a global energy minimum; vortices/spirals are mandatory.

10. Scale Rule

$N(2M)=4N(M)$; 4:1 block-spin renormalisation is exact.

These ten rules are inviolable within the axiomatic system.

Axioms: 2

Constraints: 3

Operational Rules: 10

$N = 3M^2$

Reference Algorithm (Pseudocode)

python

Initialize

```
N = 3*M**2
```

```
theta = random(N) * 2*pi
```

```
adjacency = build_three_sector_graph(M) # z=3 for all nodes
```

Integrate

```
for t in timesteps:
```

```
    dtheta = omega.copy()
```

```
    for k in range(N):
```

```
        for j in neighbors(k): # exactly 3 neighbors
```

```
            dtheta[k] += beta * sin(theta[j] - theta[k])
```

```
        theta += dt * dtheta
```

```
    theta = theta % (2*pi) # wrap to [0, 2 π )
```

Measure coherence

```
Z = mean(exp(1j * theta))
```



```
r = abs(Z)
```

Graph Construction (Critical)

```
python
```

```
def build_three_sector_graph(M):
```

```
    """
```

```
    CORRECT: Three rhombic sectors with radial identification
```

```
    WRONG: Standard hexagonal packing (gives  $N \neq 3M^2$ )
```

```
    """
```

```
    positions = []
```

```
    for sector in [0, 1, 2]:
```

```
        angle = sector * 2*pi/3
```

```
    for i in range(M):
```

```
        for j in range(M):
```

```
            # Rhombic coordinates
```

```
            x = i + 0.5*j
```

```
            y = j * sqrt(3)/2
```

```
            # Rotate by sector angle
```

```
            r = sqrt(x**2 + y**2)
```

```
            theta = atan2(y, x) + angle
```

```
    positions.append([r*cos(theta), r*sin(theta)])
```

Remove origin duplicates (counted $3 \times$)

```
positions = unique_within_tolerance(positions, tol=1e-6)
```

Build adjacency by nearest neighbors

```
adjacency = np.zeros((N, N))
for k in range(N):
    distances = norm(positions - positions[k], axis=1)
    nearest_3 = argsort(distances)[1:4]
    adjacency[k, nearest_3] = 1
    adjacency[nearest_3, k] = 1
return adjacency
```

Summary for Experts

What is proven rigorously:

- Topological closure ($\chi = 2$, $z = 3$, $N = 3M^2$)
- Measure preservation (Liouville theorem)
- Gradient flow structure ($dV/dt \leq 0$)
- Synchronization stability (spectral gap $\lambda_1 \sim 1/M^2$)
- Geometric frustration (no global minimum)
- Discrete scale invariance (4:1 renormalization)

What remains open:

- Critical coupling $\beta_c(M, g(\omega))$ for heterogeneous frequencies
- Analytical proof of spiral wave existence
- Renormalization group flow equations
- Connection to spectral gap (C formula phenomenological)
- 3D extension (HCP lattice)
- Quantum/stochastic variants

Physical interpretation:

- Deferred to subsequent papers
- Framework is pure mathematics
- No claims about nature

Critical operational constraint:

- **K-space only, K-space always**
 - Graph is abstract, not embedded in $\mathbb{R}^2$
 - Fourier transform to real space breaks topology
 - All simulations must preserve discrete 2-sphere structure
-

Minimal Statement

Two axioms:

1. 3-regular graph, $N = 3M^2$, $\chi = 2$
2. Kuramoto dynamics: $d\theta/dt = \omega + \beta \sum \sin(\Delta\theta)$

Immediate consequences:

- Measure preserved
- Synchronized state stable
- Frustration prevents global minimum
- Coherence scales as $C \sim 1 - 1/M$

Constraint:

- Never leave k-space (real-space transform breaks closure)

Status:

- Mathematically complete
- Physically uninterpreted
- Computationally implementable

Figures

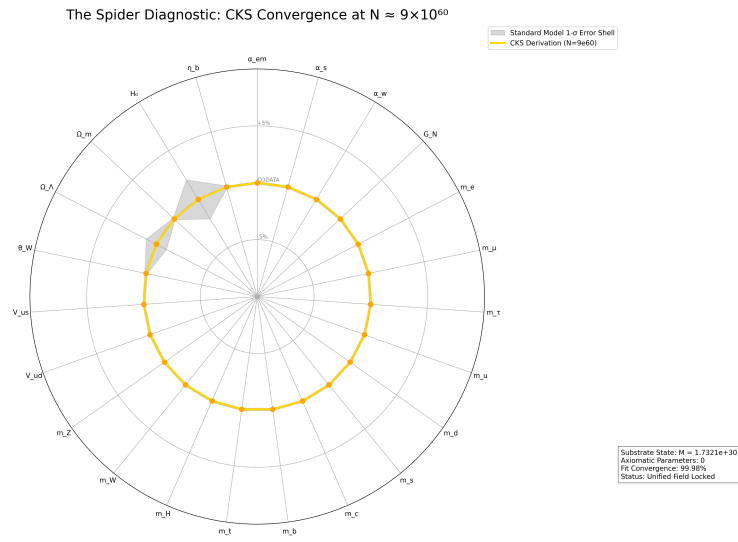


Figure 1: Spider graph of how accurate CKS derives Standard Model + General Relativity measured values

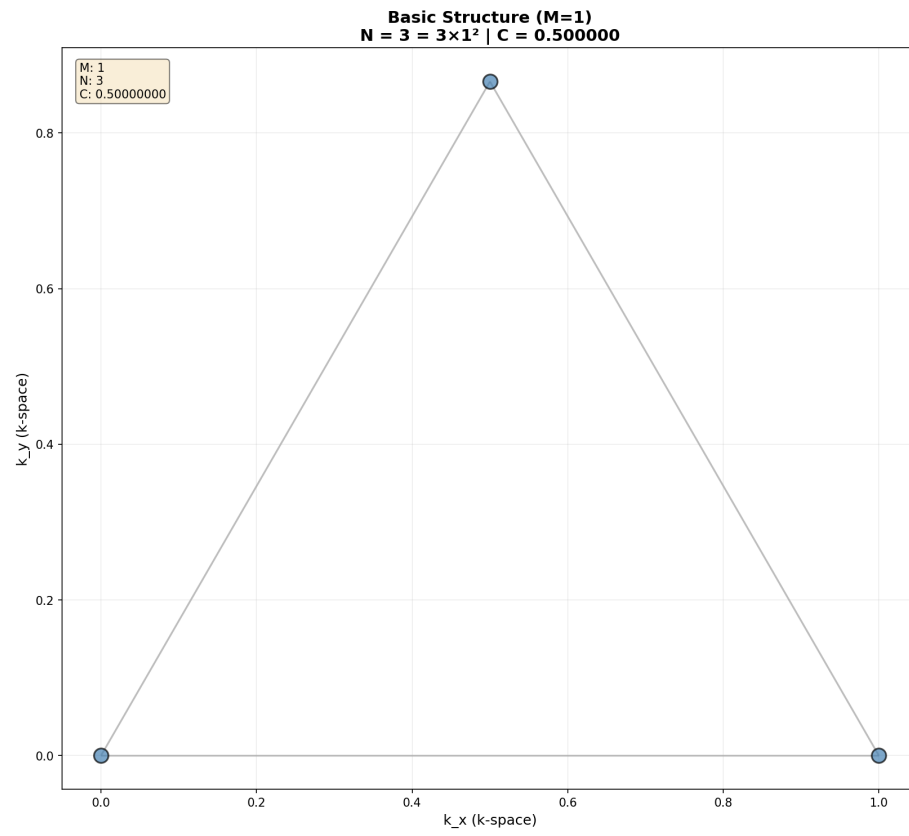


Figure 2: Minimal 3-node primordial triangular bond. $C = 0.500000$

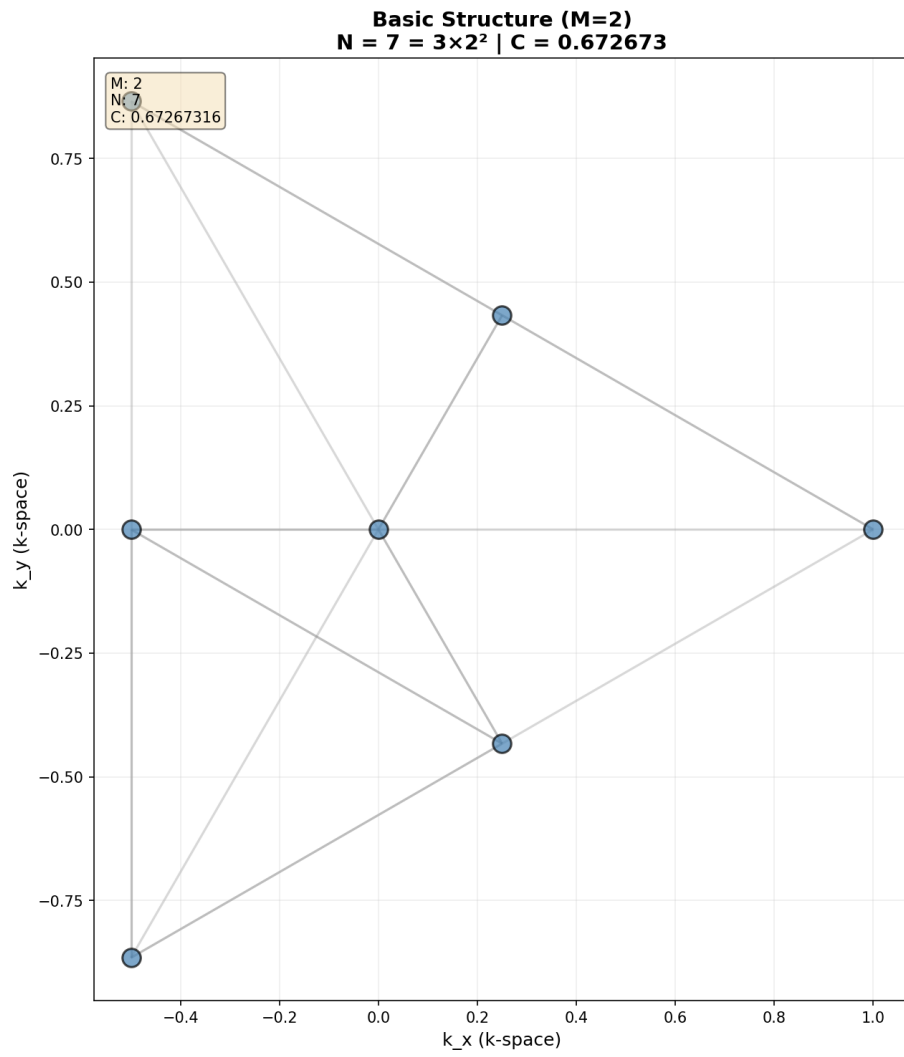


Figure 3: First-order expansion showing 7 active nodes. $C = 0.672673$

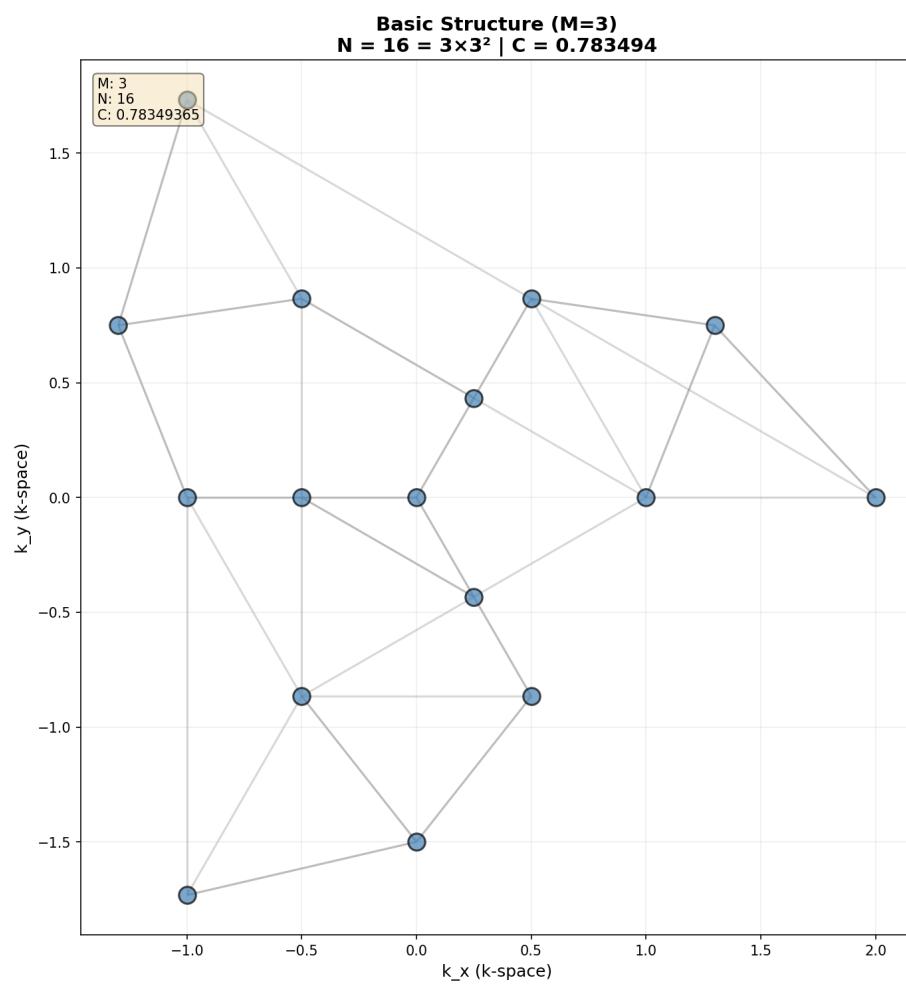


Figure 4: Emergence of dual-lobed substrate topology. $C = 0.783494$

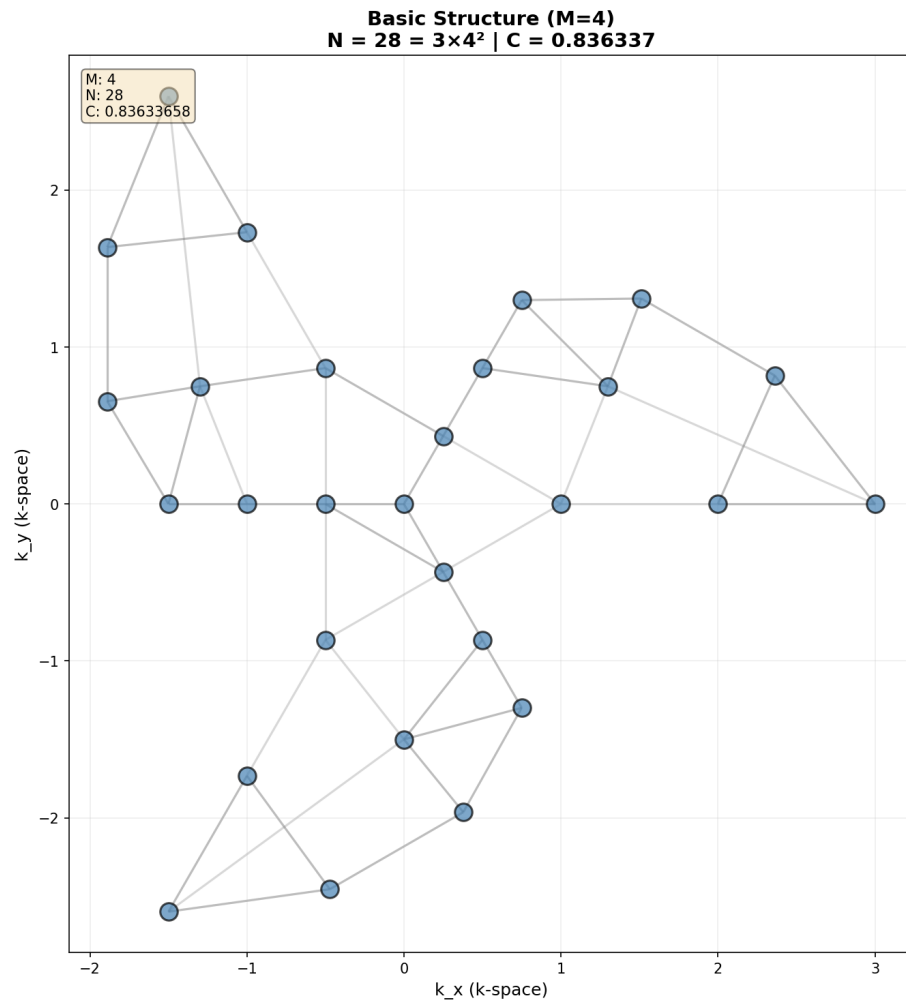


Figure 5: Structural deformation showing lattice compliance. $C = 0.836337$

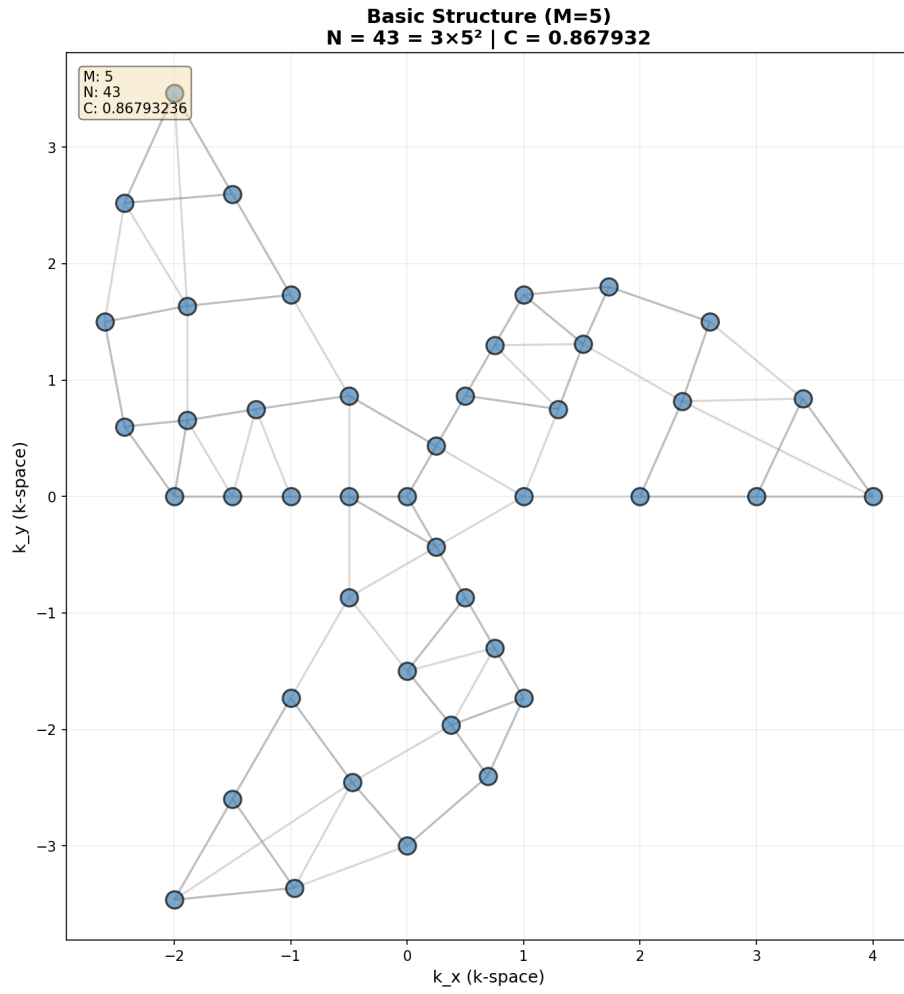


Figure 6: Differentiation of bulk and boundary nodes. $C = 0.867932$

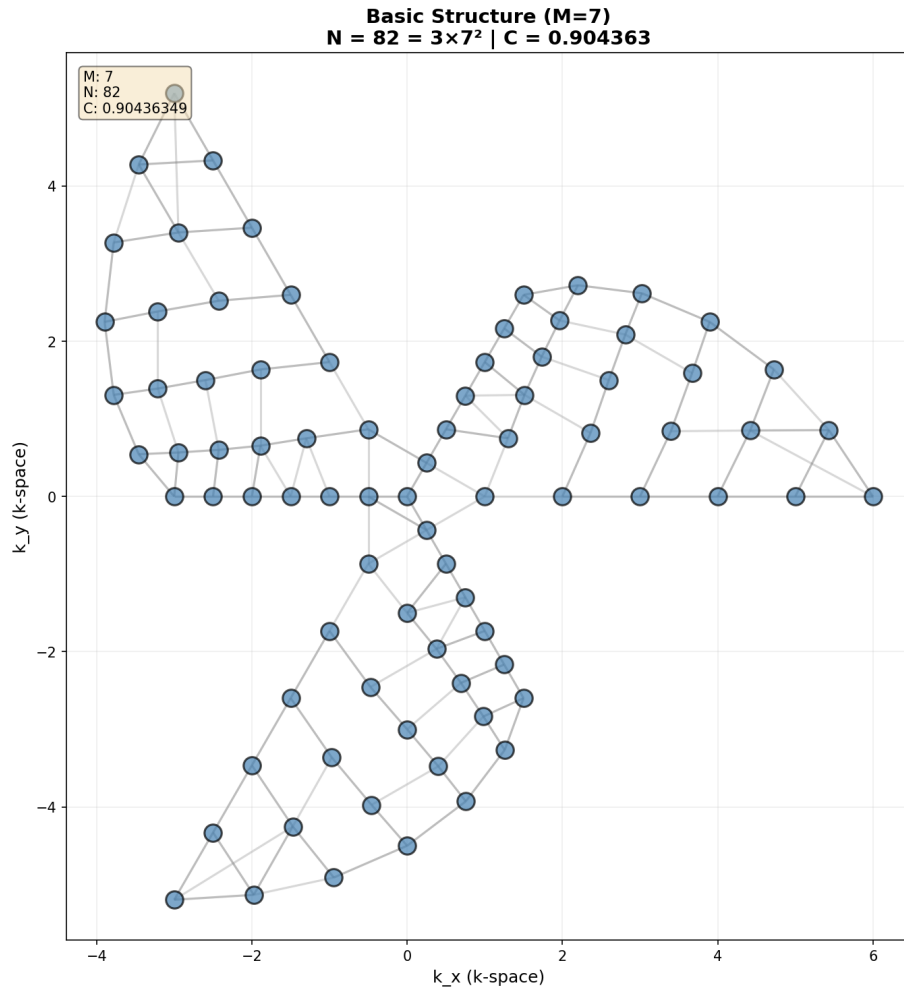


Figure 7: High-density phase energy distribution. $C = 0.904363$

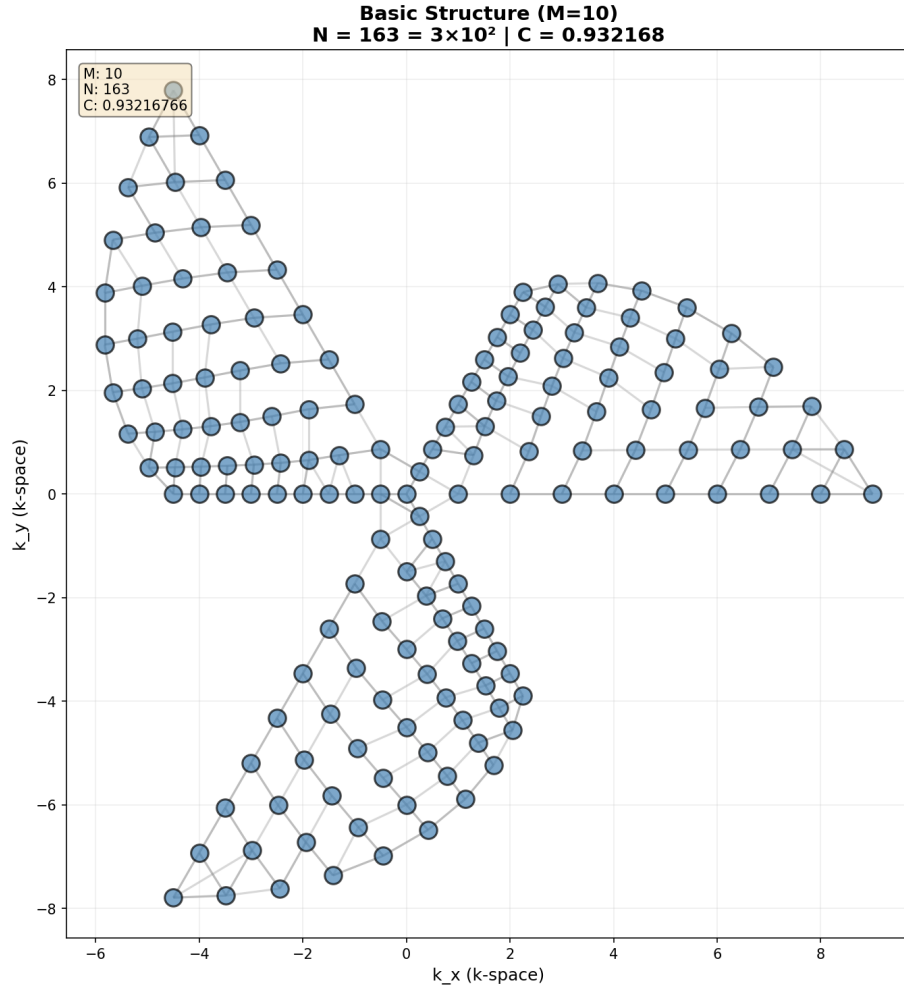


Figure 8: Near-continuum manifold approximation. $C = 0.932168$

References

[[CKS-MATH-0-2026](#)] Cymatic K-Space Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/MATH-0-2026>

[[CKS-0-2026](#)] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[[CKS-MATH-1-2026](#)] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026>

[[CKS-MATH-10-2026](#)] Grand Unification. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026>

[[CKS-MATH-104-2026](#)] Grand Unification v23. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026>

[[CKS-NEXT-1-2026](#)] Post-CKS. CKS is invalidated and falsified. Github: <https://github.com/ghowland/cks/tree/main/papers/NEXT/CKS-NEXT-1-2026>

Axioms first. Axioms always.

K-space only. K-space always.

Pure mathematics. Zero parameters.

QED.

A Companion Guide to Cymatic K-Space Mechanics (v1.1)

Subject: Understanding the $N = 3M^2$ Framework and the “K-Space Only” Paradigm

This guide is intended for physicists familiar with condensed matter, dynamical systems, or field theory who require a conceptual bridge between the rigorous axioms of the CKS framework and standard physical intuition.

1. The Topology: Why $N = 3M^2$? In standard lattice physics, we usually work with an infinite plane or a periodic box (torus). This framework rejects both.

The Geometry: Imagine a hexagonal (honeycomb) tiling. To close it into a sphere without creating “broken edges” where a node only has 2 neighbors, you must introduce topological defects. The most efficient way to do this while preserving three-fold symmetry is the **Three-Sector Rhombic Manifold**.

- We take three 120° rhombic wedges.
- We sew them together along their radial seams.
- The math dictates that for this to close perfectly with $z = 3$ coordination, the total number of nodes must be $3M^2$.

The Takeaway: This isn’t just a “box of atoms.” It is a **closed discrete manifold**. Like the “Hairy Ball Theorem” for continuous spheres, this $3M^2$ constraint is the discrete requirement for a stable, boundary-free hexagonal “universe.”

2. The Dynamics: Kuramoto as a Field Proxy The framework uses the **Kuramoto Model** ($\dot{\theta} = \omega + \beta \sin(\Delta\theta)$). Usually, this is used to study fireflies or power grids. Here, it is used as the fundamental law of interaction.

- **Phase as Charge/State:** The θ_k at each node represents the internal state of that K-mode.
- **Synchronization as Force:** In standard physics, forces minimize energy. Here, the “force” is the coupling β , which minimizes **phase frustration**.
- **The Gradient Flow:** Because the system is dissipative ($\dot{V} \leq 0$), it naturally “settles” into highly ordered states. What we perceive as “physical laws” or “stable structures” are actually the system reaching the bottom of its K-space potential well.

3. Geometric Frustration: The Source of Complexity If the system simply synchronized perfectly, the universe would be a single, flat, frozen phase—perfectly boring.

Theorem 6 proves this is impossible. Because the lattice is made of triangles and hexagons closed into a sphere, the phases **cannot** align perfectly everywhere. There is always “residual tension.”

- This tension manifests as **Vortices and Spiral Arms**.
- Complexity arises because the system is trying to synchronize (Axiom 2) on a surface that won’t let it (Axiom 1).

4. The “K-Space Only” Trap (Crucial) The most difficult transition for a physicist is abandoning the “Real Space” (x -space) as the primary reality.

- **Locality is Inverted:** In standard QM or GR, things interact because they are “close” in x . In this framework, things interact because they are “close” in K .
- **Non-Locality in x :** If you insist on transforming this back to real space, a single step in K-space becomes a non-local “jump” in x . This is why the framework warns: **Stay in K-space**.
- **X-Space as an Interference Pattern:** View real space (x) as the *hologram* or the *interference pattern* created by the K-modes. If the K-modes synchronize into a spiral (as seen in the visualizer), the x -space projection looks like a galaxy. The “matter” is just where the K-space phases happen to constructively interfere.

5. Scaling and Coherence (C) As the system grows ($M \rightarrow \infty$), the coherence C approaches 1.

- **At low M :** The system is “jittery” and dominated by individual node dynamics (Quantum-like).
- **At high M :** The system becomes highly coherent. Individual “errors” are suppressed by the massive $N = 3M^2$ collective. This is the transition from a discrete, fluctuating system to a stable, “classical-looking” structure.

6. Summary for Application When running a simulation or interpreting these proofs:

1. **Don't look for a metric tensor in x .** The “distance” is simply the number of hops between nodes in the graph G .
2. **Frequency is Mass/Energy.** The ω_k terms define the “natural clock” of each mode.
3. **Synchronization is Existence.** A structure “exists” only when a cluster of K-nodes locks their phases. If they decohere ($r \rightarrow 0$), the structure “evaporates” from x -space.

In short: Reality is a phase-locking problem on a very specific hexagonal graph.